The Data Analytics Lifecycle

DSC 340 Applied Machine Learning

7 February 2025

The Data Science Lifecycle. Lab 0 Demo. Introduce Lab 1.

Survey

- Name
- One skill you would like to develop
- One concern/worry you have about taking this class
- One thing you would like your instructor to know about you

The Plan

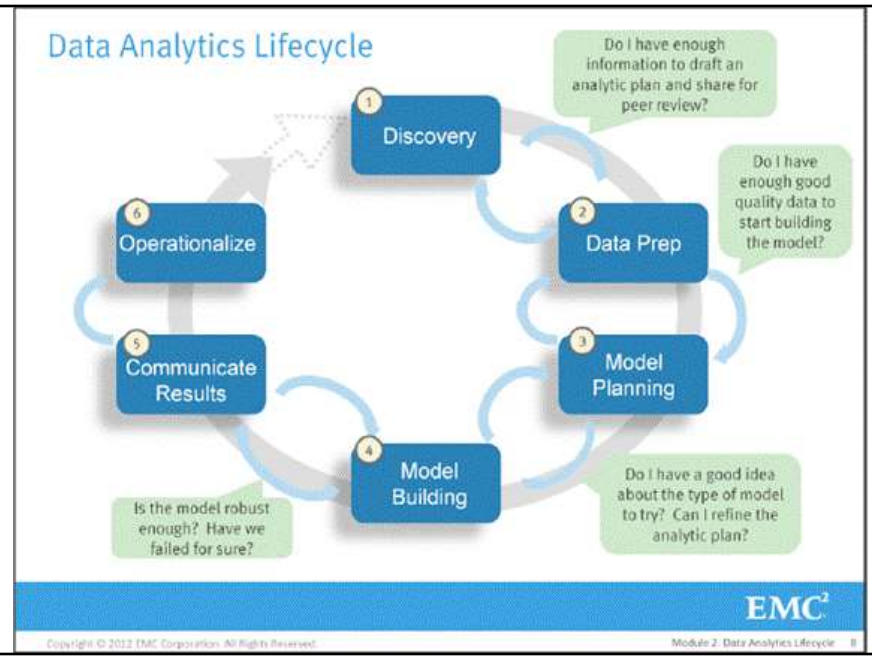
- Example projects
- Data analytics life cycle
- Lab 0: Demo and Review
- Introduce Lab 1

Example Analytics Projects

- Predict crop yields
  - Data: weather data, satellite and aerial imagery, prices, … ?
  - Challenges: Multiple crops (have to identify what is growing where!), … ?
- Increase alumni engagement
  - Data: engagement from recent years
  - Challenges: Very small database (MS Excel), incomplete information, … ?
- Predict shot success in basketball
  - Data: historic data collected by the team, videos, public databases, … ?
  - Challenges: Dynamic setting, players who come off the bench, … ?

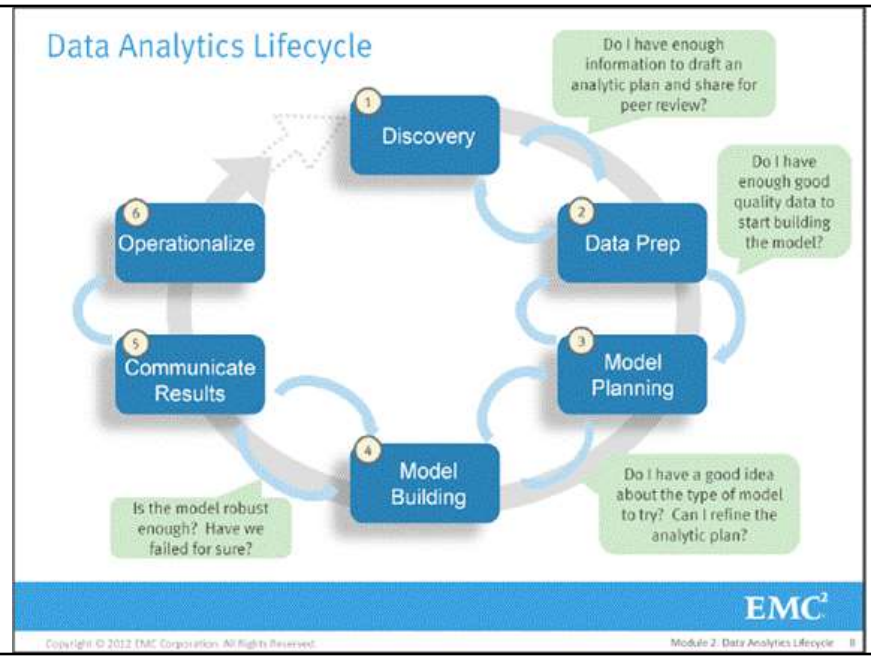
Data Analytics Lifecycle

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Data Analytics Lifecycle
Do I have enough information to draft an analytic plan and share for peer review?
1 Discovery
2 Data Prep
Do I have enough good quality data to start building the model?
3 Model Planning
Do I have a good idea about the type of model to try? Can I refine the analytic plan?
4 Model Building
Is the model robust enough? Have we failed for sure?
5 Communicate Results
6 Operationalize
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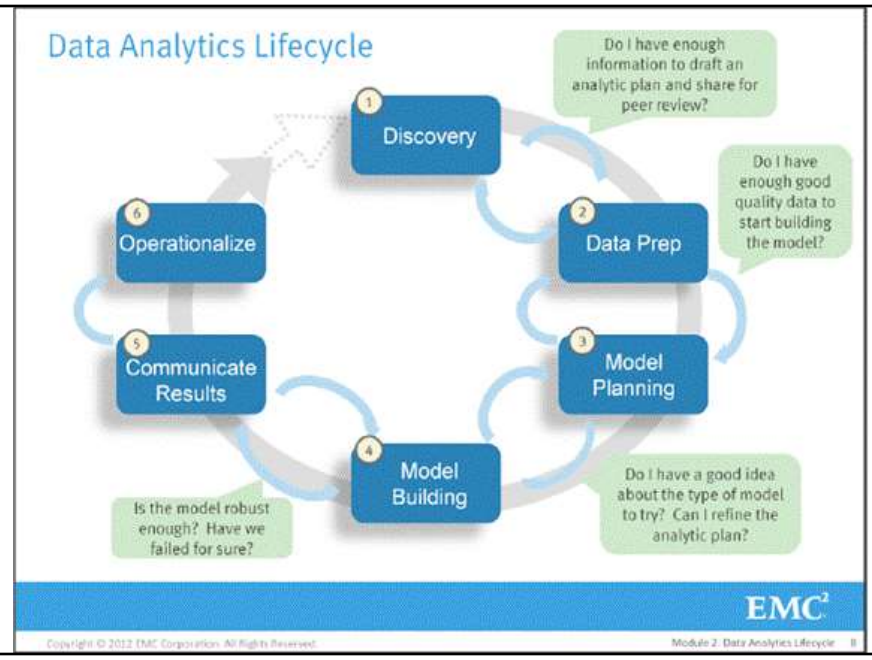
1. Discovery

- Learn the domain (business, science, etc.)
- What resources do we have? (people, tech, tools, data?)
- What problem are we solving? (frame the problem)
- Who are the stakeholders? (Who is affected? What are the risks?)
- Interview the Analytics Sponsor
- Developing initial hypotheses
- Identifying potential data sources


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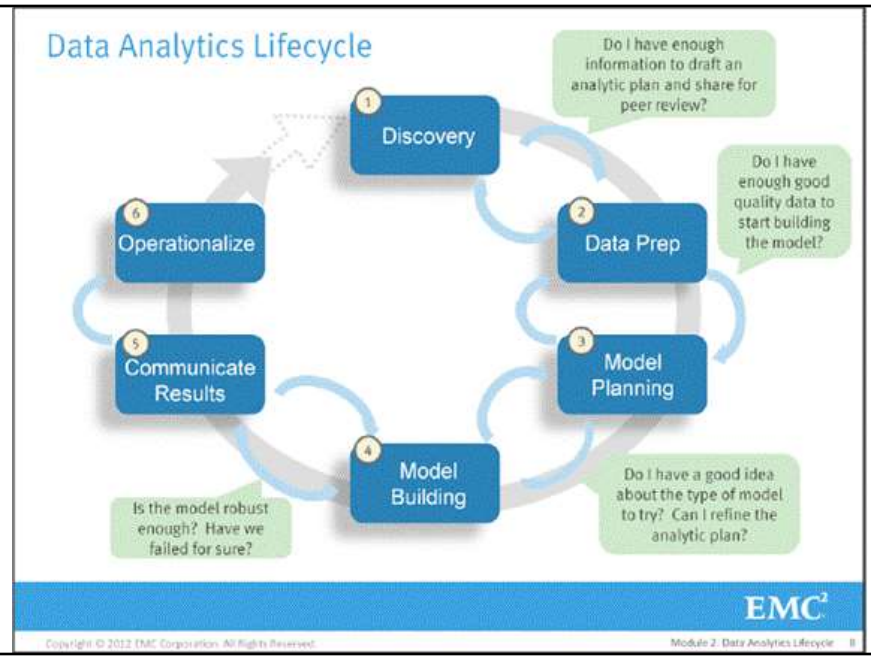
2. Data Preparation

- Sandbox data (no live databases!) analytic vs. operational DB
- Make sure you have the resources (data, tech) you need
- Learn about the data (semantics, missing values, errors, duplicates …)
- Data conditioning (address issues discovered, structure)
- Survey and visualize (is it representative? Consistent? Time frames?)


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3. Model Planning

- Data exploration and variable selection
  - Already cleaned, but need to understand relationship between variables
  - What variables should we include?
  - What variable transformations should we explore?
  - Note: model building could include choice of variables, transform method
- Model selection:
  - Is this a classification problem? Clustering?
  - Which method(s) are most appropriate?

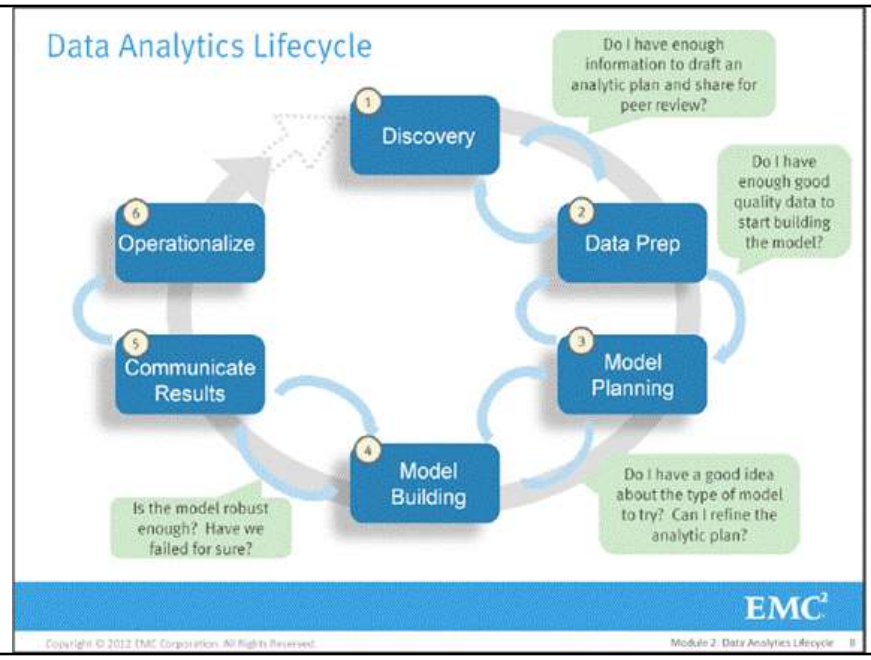

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4. Model Building

- "Train" model ("fit" data)
- Does the model appear valid and accurate?
- Does the output/behavior make sense to domain experts?
- Do the parameter values of the fitted model make sense in domain?
- Is the model sufficiently accurate to meet the goal?
- Does it avoid intolerable mistakes? (depends on context!)
  - For example: false positives. How bad?

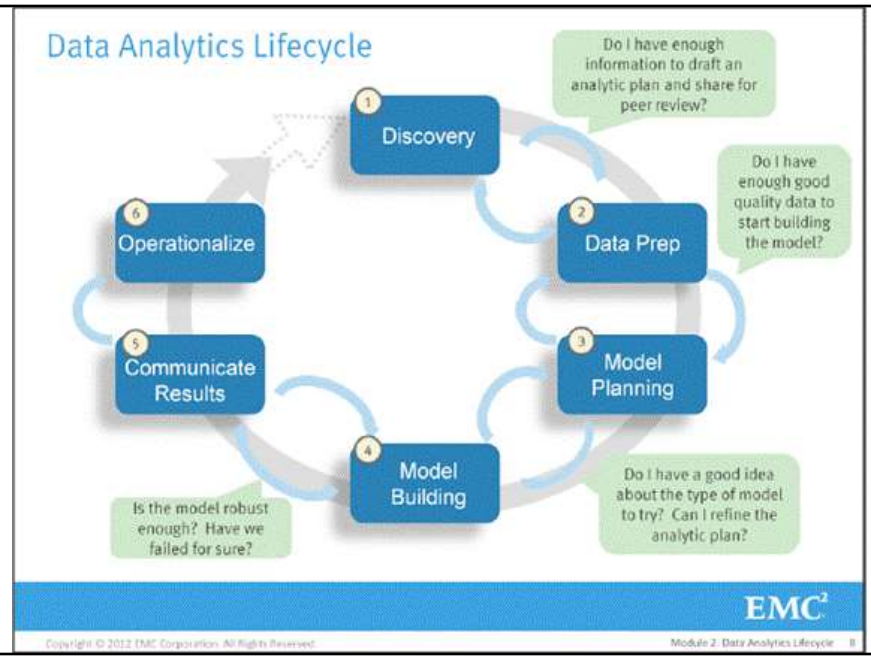
4. Model Building (cont.)

- Do we need more data? Transformations? Less data?
- Will the model support runtime (operational) requirements?
- Is a different form of the model needed for operationalization?


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5. Communicate Results

- Did we succeed or fail?
- Common sense evaluation
- Present key results
- Make recommendations for follow-up
- Document work of team


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Lab 0 Demo

The Data Analytics Lifecycle in Action

This is the pattern that we will follow especially in the later labs in this course

Introduce Lab 1

Done individually

Opportunity to review your foundational skills from CSC 170 and DSC 270

Anaconda

- We will install Anaconda on these machines
  - It will be on the machine you all are using
  - We will use my credentials to install it so all users can access it
  - You may install it to your notebook computer if you wish

Set up a work environment

- Where will your keep your files?
- Recommendation: Create a DSC340 Folder in OneDrive
- Inside that, create a Lab0 folder (or P0, etc.)
- Download the files from Moodle, put them in that directory/folder
- Inside that, create a Lab1 folder (or P1, etc.)
- Download